

ATLIQ Hardware Sales analysis project report

Atliq Hardware is a fictional computer hardware and peripheral products manufacturing company based in New Delhi, India. This project simulates the nature of an actual real world industry level data analysis project involving usage of tools like MySQL and Microsoft PowerBI for data retrieval, exploration and dashboard visualization respectively.

Business Objective

Primarily the goal of this project is to conduct Exploratory Data Analysis to extract valuable and actionable insights about the sales of the company across various parameters and features. For example, the following questions can be answered with the help of this analysis:

- 1) How do sales look like across the different markets and the calendar year?
- 2) What is the month wise revenue trends for a given year?
- 3) Which customers purchase what kind of products the most overall?
- 4) What are the key results of the firm's profit and revenue analysis?
- 5) Which retail stores were the biggest contributors in the company's sales and revenue?

Dataset Description

The dataset to be worked on was created within the MySQL workbench environment specifically inside a database named "Sales" and it contained the following tables with the respective fields: -

- **Customers:** customer_code, customer_name, customer_type
- **Date:** date, cy_date, year, month_name, date_yy_mmm
- **Markets:** markets_code, markets_name, zone
- **Product:** product_code, product_type
- **Transactions:** product_code, customer_code, market_code, order_date, sale_qty, sales_amount, currency

Data Preprocessing

In the initial stages the data tables were checked for any mistakes and inconsistencies which were found and treated accordingly. Like for this project the scope was only on the Indian markets and not foreign businesses (New York, Paris) so all respective transactions records were removed.

Within the transactions table there were some records with currency in USD i.e. US Dollars which were converted to INR using the exchange rate at the time of analysis.

Additionally, some “sales_amount” were in negative range so they were replaced with the average sales_amount in the schema.

Data Analysis

Customers

Firstly, some customer focused analysis was performed in order to find out some useful information which could be helpful in the future but mainly to understand the structure fully.

A total of **38 unique retail stores** purchased Atliq hardware products and computer components for resale to consumers. Some examples names like Premium Stores, Nixon, Modular, etc. When it comes to the type of customers there were only 2 i.e. “**Brick & Mortar**” and “**E-Commerce**”, the stores were divided in half across these types (19 each).

Overall, the number of transactions involving “Brick & Mortar” were **97,664** meanwhile for “E-Commerce” stores it was **52,619**. This tells us that “Brick & Mortar” type of customers make the higher number of sales approximately twice the amount even.

Date

The time duration covered in the dataset is for around 4 years starting from **2017** till **2020**. Specifically, from June, 2017 to June, 2020.

Markets

For the geographical information the data was divided into zones and then further into markets. A total of 3 zones were there in India which were **North**, **Central** and **South** within which multiple cities were present basically called markets.

Zone	Cities
South	5
Central	4
North	6

Zone wise number of cities (markets) present in the dataset

Some of the major cities from each zone were Chennai, Bengaluru and Hyderabad in South, Mumbai, Nagpur and Bhopal in Central and Delhi NCR, Ahmedabad in the North.

The top 3 markets that consistently recorded high sales amount and revenue for the 4 years were **Delhi NCR**, **Mumbai** and **Ahmedabad**. So, we can further investigate more deeply into these particular markets like finding out the specific stores that purchased the most from Atliq and what exact products as well.

Products

When it comes to the products that are sold by Atliq hardware there were distinctly two types which were “Own Brand” and “Distribution” which would basically mean whether the product is the company’s very own brand or is it just a distributor of that same product made by some other firm.

Product type	Product count
Own Brand	191
Distribution	88

Distribution of Product types

So, the company had more own brand products compared to distribution almost double the amount actually. Focusing on the actual number of products sold based on the type about **61,239** Own Brand products were sold on the other hand around **33,885** Distribution products were sold.

One thing to note is that depending on the product it would matter whether manufacturing you own product is better or worse than purchasing components to resell ahead which is not clear over here.

Transactions

Finally, we analyzed the transactions table that contained precisely **150,283** records with features like “sales_amount”, “sales_qty” and “currency” along with foreign keys to the previous tables in the database.

If we look at how the company performed each year then the following table does a great job depicting exactly that

Year	Total_sale (in Rs.)
2018	414322073
2019	336452114
2020	142235559
2017	93621284

After going deeper into the month wise breakdown of the total sale amount January came out to be on top with **4,25,21,659 Rs.** Then the product which showed the highest total sales in January was found out which was “Prod040” that generated about **46,28,923 Rs.** If there we look into the products that generated the most sales overall then “Prod318” leads with 6,89,67,202 Rs.

Now, if we focus on the markets recording the most total sales then “**Delhi NCR**”, “**Mumbai**” and “**Ahmedabad**” consistently place in the top 3 of each year. After digging deeper to determine which store purchased the most goods it turned out to be “**Premium Stores**” which came up about **19,823** times in total.

The following table shows the product code, type and the frequency for “Premium Stores”: -

product_code	product_type	product_count
Prod252	Distribution	517
Prod102	Own Brand	504
Prod271	Own Brand	450
Prod116	Distribution	416
Prod129	Distribution	399

Key Findings & recommendations

Based on our analysis Atliq Hardware can focus on the suggested points for improvements and benefits for the future.

1. Maintain emphasis on “Brick & Mortar” primarily as customers rely on physically purchasing hardware products as opposed to online but they can do research on other E-commerce websites and the customers that have been relatively successful in selling computer hardware parts to see what more can be done.
2. Consider partnering with more local retailers in the Southern zone to bring the sales up in the respective markets that were in the bottom ranks in terms of sales and purchase frequency.
3. Continue investing in “Own Brand” product types and then conduct more research on “Distribution” products to figure out what improvements can be made.